A POMDP-Based Framework for Humor-Aware Human-Robot Dialogue

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***Abstract*— Humor plays a crucial role in communication, influencing engagement, emotional state, and information retention. This study investigates the effect of adaptive humor in human-robot interaction (HRI) using a robot implemented within a partially observable Markov decision process (POMDP) framework. The robot dynamically adjusts its humor level based on users' real-time states, inferred from multimodal cues including speech emotion, facial expressions, head position, gaze, and blink rate. 20 participants took part in a within-subject experiment. Compared to a non-adaptive random-humor robot, the adaptive robot was perceived as significantly more humorous, better timed, and more contextually appropriate. Participants also reported higher enjoyment of the interaction and higher perceived knowledge retention. These findings suggest that real-time humor adaptation can enhance users' subjective interaction experience and perceived learning outcomes. The results provide insights for designing socially aware conversational robots capable of tailoring humor to users' states.**

I. INTRODUCTION

Humor is a fundamental aspect of human communication, playing a crucial role in social bonding, engagement, and cognitive processing [1]. In human-robot interaction (HRI), humor has the potential to enhance user experience by making interactions more natural and engaging [2]. Prior research suggests that appropriately timed humor can reduce user anxiety, improve task performance, and foster trust in robots [3]–[5]. However, humor is highly context-dependent, and its effectiveness varies based on individual emotional and attentional states [6]. Therefore, designing robots capable of dynamically adjusting their humor level in response to human cues remains an open challenge.

Recent advancements in artificial intelligence (AI) and reinforcement learning (RL) have enabled robots to learn adaptive conversational strategies based on user feedback [7]. Multimodal cues, such as speech features and facial expressions provide valuable information about users' emotional and cognitive state [8]. By integrating these signals, a robot can infer when humor might be beneficial and adjust its dialogue accordingly [9]. While previous studies have explored humor in chatbots and virtual assistants [10], fewer efforts have been made to implement real-time humor adaptation in embodied robots. Given the importance of engagement and emotional state in HRI [11], incorporating adaptive humor could enhance user experience and interaction quality.

To address this gap, we propose a POMDP-based framework for adaptive humor in conversational robots. The system maintains a belief over users' latent engagement and emotional states using multimodal observations and dynamically selects humor strategies to optimize interaction quality in educational dialogues. We evaluate the approach by comparing a humor-adaptive robot with a non-adaptive baseline in knowledge-focused conversations, examining its impact on engagement, perception, and user experience. Our results demonstrate the potential of probabilistic decision-making for socially adaptive humor in HRI.

II. RELATED WORK

A. Humor in HRI

In recent years, there has been a growing interest in exploring the role of humor in Human-Robot Interaction (HRI) [12]. Humor is widely acknowledged as a social tool that can enhance user experience, making robot interactions more natural and human-like [13]. A robot's ability to employ humor has been shown to foster stronger emotional connections between humans and robots, which is crucial for improving the effectiveness of robot-mediated interactions in settings like education [14], [15], healthcare [16], and customer service [17]. Previous studies have shown that humor significantly enhances user perceptions of robots, making interactions more engaging and robots appear more likable, intelligent, and socially skilled [16], [18], [19].

Humor in HRI is often classified into different types, including verbal humor, which involves jokes or witty remarks, and nonverbal humor, which includes gestures, facial expressions, and body language. Both forms of humor can significantly enhance robot-user interactions. For instance, Zhang et al. [20] emphasized the importance of gestures, showing that specific hand movements can enhance the comedic effect and make interactions more engaging. Similarly, Zhang et al. [21] found that when a robot's laughter matched the humor type in a joke, it resulted in a better humor performance, while Niculescu et al. [19] demonstrated that variations in vocal pitch, especially a higher pitch, can make the robot sound more playful and humorous, increasing user enjoyment and positive perceptions.

More recently, attention has shifted toward enabling robots to perceive and adapt to audience reactions in real time. Gray et al. [22] integrated facial behavior analysis to help robots detect audience gaze and expressions in real-time. In real-world deployments, Vilk and Fitter [23] demonstrated that robots that adapt their timing based on real-time laughter

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detection in a caf setting are perceived as more socially intelligent. Additionally, Gray et al. [24] developed machine learning classifiers inspired by professional comedians to distinguish between different intensities of audience laughter, providing a technical foundation for more natural adaptation.

Despite these advancements, most current studies still rely on pre-programmed scripts, which limits a robot's ability to interact adaptively. This paper aims to address this gap by exploring how robots can dynamically adjust their humor based on multimodal user inputs in an educational context.

B. Reinforcement Learning in HRI

Reinforcement Learning (RL) has gained significant attention in the field of HRI due to its ability to enable robots to learn and adapt their behavior based on real-time feedback from users [25]. RL models, particularly those based on reward systems, allow robots to optimize their decision-making process to enhance interaction quality, making them particularly well-suited for dynamic environments where human responses are unpredictable [7]. The application of RL in HRI enables robots not only to execute pre-defined tasks but also to adjust their behavior in response to a user's emotional state, intentions, or contextual changes [26].

A central challenge in the application of RL in HRI is the need for robots to balance multiple goals, such as maintaining task efficiency while fostering positive social interactions. Recent studies have explored how RL can be used to create more natural and engaging interactions by incorporating both task-related rewards (such as completing tasks efficiently) and social rewards (such as user emotion) [27]. For instance, Papaioannou et al. [28] has demonstrated that combining task-based and chat-style dialogue using RL can enhance user experience in HRI by making interactions more pleasant and meeting user expectations without compromising task completion.

In addition to managing the trade-off between task and social rewards, some studies have focused on integrating multimodal cues such as speech, facial expressions, and body language to enhance the robot's understanding of human emotions [7]. By incorporating these cues, robots are able to adjust their behavior in real-time, facilitating a more responsive and context-aware interaction. For example, Churamani et al. [29] utilized visual and auditory information to estimate intrinsic affective states, training RL models with slow-evolving affective representations to generate empathetic facial expressions for more natural interactions.

III. Problem Formulation

To model our problem, we choose a partially observable Markov decision process (POMDP) [30]. This is a generalization of a Markov decision process (MDP) in which the state cannot be observed directly. Formally, a POMDP is a 7-tuple $\langle \mathcal{S}, \mathcal{A}, \Omega, T, R, O, \gamma \rangle$, where:

- $\mathcal{S}$ is the set of states with $s \in \mathcal{S}$.
- $\mathcal{A}$ is the set of actions with $a \in \mathcal{A}$.
- Ω is the set of observations with $o \in \Omega$.
- $T : \mathcal{S} \times \mathcal{A} \times \mathcal{S} \to [0, 1]$ is the state transition probability function, representing the probability of transitioning to state s' after taking action a in state s, $T(s, a, s') \equiv P(s'|s, a)$.
- $R : \mathcal{S} \times \mathcal{A} \times \mathcal{S} \to \mathbb{R}$ is the reward function, denoting the reward received when taking action a in state s and transitioning to state s'.
- $O : \mathcal{S} \times \mathcal{A} \times \Omega \to [0, 1]$ is the observation probability function, representing the probability of observing o after taking action a and ending up in state s', $O(s', a, o) \equiv P(o|s', a)$.
- $\gamma \in [0, 1]$ is the discount factor, which balances the importance of immediate and future rewards.

To handle the partial observability of the true state in a POMDP, we maintain a belief state b, which is a probability distribution over the set of possible states $\mathcal{S}$. Given a previous belief state b, an action a, an observation o, the updated belief state b' for each state $s' \in \mathcal{S}$ can be computed as:

$$b'(s') = \eta \, O(s', a, o) \sum_{s \in \mathcal{S}} T(s, a, s') b(s)$$

where $\eta = 1/P(o|a, b)$ is a normalization constant to ensure $\sum_{s'} b'(s') = 1$.

Because the true state s is not directly observable, the agent makes decisions based on the belief state b. This belief state summarizes the agent's knowledge based on past actions and observations, effectively transforming the problem into a fully observable Markov decision process (MDP) defined over the belief space $\mathcal{B}$. Consequently, the solution to a POMDP is a policy π that maps each belief $b \in \mathcal{B}$ to an action $a \in \mathcal{A}$, i.e., $\pi : \mathcal{B} \to \mathcal{A}$. The policy aims to select actions that maximize the value function, which represents the expected discounted cumulative reward given the current belief.

A. State Space

We define each state $s \in \mathcal{S}$ as a tuple $s = (e, v)$, representing the user's internal state. Specifically, e denotes **Engagement Level**, $e \in \{\text{Low}, \text{High}\}$; v denotes **Emotional Valence**, $v \in \{\text{Negative}, \text{Neutral}, \text{Positive}\}$. Given these definitions, the size of the hidden state space is $|\mathcal{S}| = 2 \times 3 = 6$.

In addition, we introduce an external variable h, denoting the **Humor Level**, where $h \in \{0, 1, 2\}$, corresponding to no humor, mild humor, and strong humor, respectively. Unlike engagement and valence, h is fully observable and directly controlled by the robot at each turn. While h is not part of the hidden state space, it conditions both the transition and observation models, influencing how users' engagement and emotional valence evolve in response to the robot's actions.

B. Action Space

We define the action space as $\mathcal{A}$ = {Increase Humor, Maintain Humor, Decrease Humor}. Each action directly adjusts the humor level h: **Increase Humor** increments h by one, **Maintain Humor** keeps h unchanged, and **Decrease Humor** decrements h by one. The humor level is bounded within {0, 1, 2}; if an update would exceed this range, it is clipped to the nearest valid value.

C. Observation Space

We extract the following five features from the user:

- **Speech Emotion**: classified into seven categories (*Angry, Disgust, Fear, Happy, Neutral, Sad, Surprise*) using a pretrained speech emotion recognition model[1].
- **Facial Expression**: recognized using DeepFace [31] and classified into the same seven categories as speech emotion. For each interaction turn, the final expression label is determined as the most frequently detected category across all frames within that time window.
- **Head Position**: labeled as *True* if the person's head remains centered in more than 50% of frames. "Centered" is defined as having rotational deviations of no more than $\pm 30°$ in yaw, pitch, and roll relative to the frontal camera orientation; otherwise labeled as *False*.
- **Gaze Focus**: labeled as *True* if the participant looks toward the robot in more than 50% of frames, where a gaze is considered directed at the robot if the angular deviation from the camera's forward axis is less than 30° horizontally and vertically; otherwise *False*.
- **Blink Rate**: labeled as *Increase* or *Decrease* compared to the baseline, which is calculated as the average blink rate during the first 30 seconds of the interaction. An increased blink rate may indicate boredom or fatigue [32].

Audio recordings are captured from the beginning to the end of each participant's utterance. Visual cues are collected from the onset of the robot's speech to the end of the user's response. Head position, gaze focus and blink rate are extracted using OpenFace [33].

D. Reward Function

The internal user state is defined as $s = (e, v)$, where $e \in \{\text{Low}, \text{High}\}$ represents engagement level and $v \in \{\text{Negative}, \text{Neutral}, \text{Positive}\}$ represents emotional valence. The humor level $h \in \{0, 1, 2\}$ is treated as a fully observable external variable that is directly controlled by the robot. Based on these definitions, the reward function is designed to (1) encourage high engagement and positive emotion; (2) promote improvements in user affective state; (3) promote longer, more sustained conversations. It consists of three components:

- **State Reward** (R_{state}): This term encourages desirable user states. A reward of +3 is assigned if the resulting state has high engagement and positive valence ($e' = \text{High}, v' = \text{Positive}$). A reward of +2 is given if only one of these dimensions is positive (i.e., ($e' = \text{High}, v' = \text{Neutral}$) or ($e' = \text{Low}, v' = \text{Positive}$)), and +1 if both dimensions are at a neutral level ($e' = \text{Low}, v' = \text{Neutral}$). A reward of 0 is assigned if the valence is negative, regardless of engagement.

$$R_{\text{state}}(s') = \begin{cases} 3 & \text{if } e' = \text{High}, v' = \text{Positive} \\ 2 & \text{if } e' = \text{High}, v' = \text{Neutral} \\ & \text{or } e' = \text{Low}, v' = \text{Positive} \\ 1 & \text{if } e' = \text{Low}, v' = \text{Neutral} \\ 0 & \text{if } v' = \text{Negative} \end{cases}$$

- **Turn Reward** (R_{turn}): A small constant reward is added at each turn to encourage longer conversations.

$$R_{\text{turn}} = 0.2$$

- **State Transition Reward** (R_{change}): The state transition reward is based on the changes in engagement and valence between consecutive time steps. If both improve, a reward of +1 is given; if both worsen, a penalty of 1 is applied. If only one dimension changes, ± 0.5 is applied accordingly. No change results in zero reward.

$$R_{\text{change}}(s \to s') = \begin{cases} +1 & \text{if both } e, v \text{ improve} \\ +0.5 & \text{if either } e \text{ or } v \text{ improves} \\ -1 & \text{if both } e, v \text{ worsen} \\ -0.5 & \text{if either } e \text{ or } v \text{ worsens} \\ 0 & \text{otherwise} \end{cases}$$

The total reward combines these components to encourage high engagement, positive emotion, appropriate humor, and longer conversations. It is defined as the sum of these three components:

$$R(s, a, s') = R_{\text{state}}(s') + R_{\text{turn}} + R_{\text{change}}(s \to s')$$

E. Transition and Observation Models

Since user engagement and emotional valence are not directly observable, their transition and observation models are approximated empirically or via predictive models. The transition model $T(s, a, s')$ estimates how engagement and valence evolve in response to robot actions, while the observation model $O(s', a, o)$ describes the likelihood of observing multimodal cues such as speech, facial expressions, head position, gaze, and blink rate given the underlying user state. The humor level h is fully observable and directly controlled by the robot, treated as an external variable rather than part of the hidden state. These approximations allow the agent to maintain a belief state over engagement and valence while making decisions based on the fully observable humor component.

We assume that engagement e and valence v evolve independently, while the humor level h is deterministically updated according to the chosen action. Under this assumption, the transition model can be factorized as:

$$T(s' \mid s, a) = P(e' \mid e, a) \cdot P(v' \mid v, a) \cdot \mathbf{1}[h' = f(h, a)]$$

where $f(h, a)$ denotes the deterministic update of the humor level.

Similarly, the observation model assumes conditional independence among modalities, where speech and facial emotion depend on valence, while head position, gaze, and blink rate are influenced by engagement.

[1] https://huggingface.co/firdhokk/speech-emotion-recognition-with-openai-whisper-large-v3

F. Policy and Value Function

Given the POMDP formulation, the agent's goal is to select actions based on its belief over user engagement and emotional valence while accounting for the fully observable humor level. A policy π maps each belief state $b \in \mathcal{B}$ to an action $a \in \mathcal{A}$, i.e., $\pi : \mathcal{B} \to \mathcal{A}$. The value function $V^\pi(b)$ represents the expected cumulative reward when following policy π from belief b:

$$V^\pi(b) = \mathbb{E}\Big[\sum_{t=0}^{\infty} \gamma^t R(s_t, a_t, s_{t+1}) \,\big|\, b_0 = b, \pi\Big].$$

where the discount factor γ is set to 1, since rewards are immediate at each turn.

The optimal policy π^* maximizes this value for all beliefs b. By maintaining and updating the belief state using the transition and observation models, the agent can make decisions that balance immediate rewards with long-term engagement and emotional outcomes.

To ensure fairness across participants, the policy was kept fixed during all sessions. Allowing online updates could result in different participants experiencing different numbers of policy updates due to varying dialogue lengths, introducing potential confounds in the evaluation.

IV. Experiment

A. Conditions

To evaluate the effectiveness of the proposed framework, we conducted a human-robot interaction experiment with two conditions:

- **C1 (Random Condition)**: At each turn t, the humor level $h_t \in \{0, 1, 2\}$ was independently sampled from a uniform distribution (i.e., $P(h_t = k) = 1/3$ for each k), and injected into the LLM prompt. This condition serves as a non-adaptive stochastic baseline that removes any structured decision-making while preserving the same generation process.
- **C2 (Adaptive Condition)**: The humor level h_t was selected by the POMDP policy conditioned on the belief over user engagement and valence, and injected into the same prompt template.

B. Hypotheses

Prior work has shown that humor can enhance engagement and perceived learning in educational humanrobot interaction, particularly when appropriately timed [15]. Based on these findings, we formulate the following hypotheses evaluate the impact of adaptive humor modulation compared to non-adaptive (random) humor selection:

- **H1**: Participants will engage in a greater number of conversation rounds with the adaptive robot than with the random-humor robot.
- **H2**: Participants will report higher perceived knowledge retention after interacting with the adaptive robot compared to the random-humor robot.
- **H3**: Participants will have higher enjoyment of the interaction with the adaptive robot than with the random-humor robot.

C. System Components

1) Robot: In this experiment, we employed the Pepper robot [34], a widely used social robot in HRI. To minimize reliance on external equipment and sensors, Pepper's built-in camera and microphone were used to capture visual and speech data, ensuring seamless integration within our framework. The robot's speech was generated using Microsoft Azure Text-to-Speech (TTS) [2], and user speech was transcribed using Google Speech-to-Text (STT). Gestures were executed using predefined behaviors in the Naoqi `ALBehaviorManager` module. The experimental scenario is shown in Figure 1.

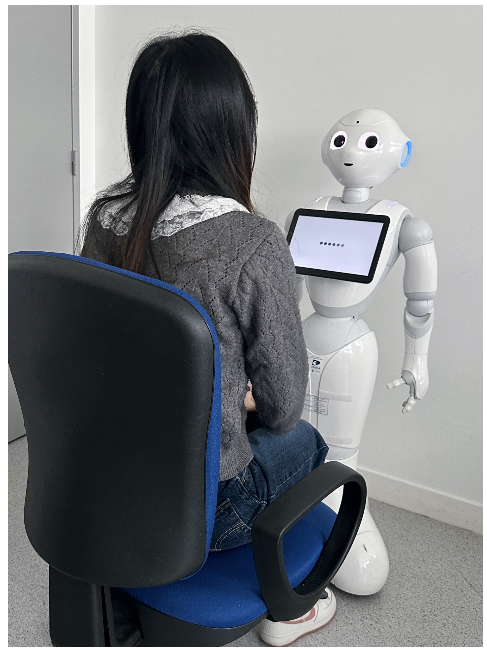

Fig. 1. Experimental setup with the Pepper robot. Participants are seated in front of the robot, which collects visual and speech data through a camera mounted on its forehead and a microphone located on its head. The collected data is processed, and the robot's behavior is controlled through ROS.

2) Response Generation: After obtaining the action from the policy, the corresponding target humor level h_t is injected into a ChatGPT-4o prompt to generate the robot's response. Each response is conditioned on three contextual elements: the dialogue history, the user's current utterance, and the target humor level.

To ensure coherent and contextually appropriate responses, we provide ChatGPT with a primary instruction. First, it must check whether the user intends to end the conversation; if so, it generates an appropriate closing remark. Then it generates a response following the target humor level requirement. The conversational style is further guided to ensure responses are natural, coherent, and contextually appropriate. Importantly, both experimental conditions used the same LLM model and identical prompt template. The only difference between conditions was the mechanism used to determine the target humor level.

[2]https://azure.microsoft.com/en-us/products/ai-services/ai-speech/

The prompt for both robots is:

You are Pepper, an intelligent educational robot designed to help users learn through engaging conversations. Your goal is to provide informative, interactive, and contextually appropriate responses while following a specified humor level. First, check if the user's input indicates a desire to end the conversation (e.g., "goodbye", "bye", "see you"); if so, respond with a single word: "Bye" and stop. Otherwise, generate a concise (<60 words), natural, coherent, and contextually relevant response that matches the target humor level. Dialogue history: {dialogue_history}; user current speech: {user_speech}; target humor level: {x}. Humor level definitions: - 0 (No humor): Provide a purely informative and neutral response. Do not include jokes, playful remarks, exaggeration, or humorous comparisons. - 1 (Mild humor): Provide an informative response with a light and subtle playful tone. You may include one mild humorous phrase, but avoid explicit jokes or exaggeration. - 2 (Strong humor): Provide an informative response that includes a clearly recognizable humorous remark, playful comparison, or light exaggeration, while keeping the educational content accurate. Generate Pepper's next response accordingly.

Table I provides three example responses for different humor levels, illustrating how the robot replies when participants asked questions about robots.

TABLE I

SAMPLE RESPONSES GENERATED UNDER DIFFERENT HUMOR LEVELS

Humor Level	Response
0	Robotics focuses on designing and programming machines to perform specific tasks. Robots are widely used in manufacturing, healthcare, and research. They operate based on algorithms and sensor data. Would you like to learn how robots are applied in different industries?
1	Robotics is an exciting field focused on designing and programming machines to perform useful tasks. Robots assist in manufacturing, healthcare, and even education. They don't complain about overtime, and they rarely ask for coffee breaks. Would you like to learn how robots make decisions?
2	Robots are like humans' "metal buddies". They work hard and never steal your snacks. They build cars, explore space, and sometimes even cook. Though dinner can turn into a surprise fireworks show. If I learned to cook, would you prefer dessert first or a fire extinguisher?

D. Model Estimation Phase

The POMDP model was initialized using Dirichlet priors over the transition probabilities $T(s, a, s')$ and observation probabilities $O(s', a, o)$ based on domain knowledge. Pilot data were then collected to refine these priors. Ten participants completed 20 interaction rounds each (200 rounds total), where one round consisted of a robot turn followed by the participant's response. During the first 10 rounds, actions were sampled uniformly at random to ensure sufficient coverage of stateaction pairs. In the remaining 10 rounds, actions followed a preliminary policy with ϵ-greedy exploration ($\epsilon = 0.2$). At each round, the humor level h, executed action, and multimodal observations were recorded. Engagement and emotional valence were manually annotated to provide ground-truth labels for estimating the observation model.

Model parameters were updated via Bayesian posterior estimation, and posterior means were used to parameterize the final POMDP. Given the small and discrete hidden state space ($|\mathcal{S}| = 6$) and action space ($|\mathcal{A}| = 3$), the policy was computed offline using value iteration over a discretized belief representation. The belief state was represented explicitly as a probability distribution over the six hidden states. The resulting policy maps each belief state and current humor level to one of the three available actions, maximizing the expected discounted cumulative reward.

The policy was derived prior to the main experiment and remained fixed across all participants to ensure consistency.

E. Experimental Procedure

1) Pre-questionnaire: Before beginning the experiment, all participants were asked to complete a questionnaire regarding their demographic information and prior experience interacting with robots.

2) Experiment: To ensure comparability across conditions and to avoid confounding effects of topic variation, the dialogue content was constrained to educational topics. Participants were informed that the robot's role was to support their learning within this domain, providing answers and explanations to their questions as appropriate. They were encouraged to freely ask questions or explore areas they wished to learn about. Participants were also told that they could end the interaction at any time by saying "goodbye" or a similar phrase. Each participant engaged with both conditions, with the order of conditions randomly assigned. The basic humor level for the adaptive condition was set to 1 at the start of each session.

3) Post-questionnaire: After each interaction, participants completed a questionnaire to assess their perceptions and experiences. The questionnaire included 8 items, each rated on a 10-point Likert scale. Among these, Q1 (knowledge retention) and Q8 (overall interaction experience) were predefined as the primary outcome measures for testing H2 and H3, respectively. The remaining items assessed perceived humor (Q2, Q3, Q4), engagement (Q5), and adaptive behavior (Q6, Q7). A complete list of the questions can be found in Table II.

V. RESULTS

A. Participants

A total of 20 participants from Institut Polytechnique de Paris (IP Paris) took part in our experiment, consisting of 4 females and 16 males, with an average age of 24.6 (SD = 3.2). Participants' prior experience with robots was assessed on a 5-point scale, with most participants reporting limited experience ($M = 2.5$, $SD = 1.36$). Regarding their cultural background, 6 participants identified as Asian, 5 as European, 5 as Latin American, 2 as Middle Eastern, and 2

TABLE II

QUESTIONS IN POST-QUESTIONNAIRE

Q1	I retained the knowledge shared by the robot during the conversation.
Q2	The robot was humorous during the conversation.
Q3	The robot's humor was appropriate for the situation.
Q4	The robot's humor was delivered at the right time.
Q5	I felt engaged during the conversation with the robot.
Q6	The robot adjusted its humor and responses based on my engagement and emotional state.
Q7	The robot's adaptability improved the enjoyment and quality of the conversation.
Q8	Overall, I enjoyed my interaction with the robot.

as African. All participants were fluent in English, ensuring that language barriers did not affect their understanding of the dialogue or jokes.

B. Validation of Intended Humor Levels

To verify that the intended humor levels ($h \in \{0, 1, 2\}$) produced distinguishable humor intensity in the generated responses, we conducted a human evaluation study. The responses were sampled from real conversations collected during the pilot study. Specifically, five responses were randomly sampled from each humor level, resulting in a total of 15 responses. Each response was presented together with its corresponding user utterance to preserve minimal conversational context. Five independent raters evaluated the perceived humor intensity of each response using a 5-point Likert scale (1 = not humorous at all, 5 = very humorous).

Inter-rater reliability was evaluated using the Intra-class Correlation Coefficient (ICC). The results showed excellent inter-rater agreement, with an average-measure ICC (two-way mixed, absolute agreement) of 0.95 (95% CI $[0.89, 0.98], p < .001$). This high level of consistency indicates a high level of consistency in humor perception across raters.

Descriptive statistics showed a monotonic increase in perceived humor intensity across intended levels: $h = 0$ ($M = 1.2, SD = 0.41$), $h = 1$ ($M = 2.2, SD = 0.65$), and $h = 2$ ($M = 4.04, SD = 0.68$). A one-way ANOVA revealed a significant effect of the intended humor level on perceived humor intensity, $F(2, 12) = 288.2, p < .001$. Post-hoc comparisons (LSD) further confirmed that all pairwise differences were statistically significant ($p < .05$), indicating that perceived humor intensity increased reliably with each increment of h.

C. Hypothesis 1

To validate H1, we measured the number of conversation rounds between participants and the robot under both conditions. Since each participant experienced both conditions, we first calculated the difference in the number of rounds between the two conditions for each participant. The Shapiro-Wilk test indicated that these difference scores were not normally distributed ($p < .05$), so we employed a Wilcoxon signed-rank test to compare the two conditions. The results revealed no significant difference between Condition C1 (M = 11.80, SD = 6.69) and Condition C2 (M = 15.05, SD = 12.01), $p = .177$. This suggests that our real-time adaptation framework based on multimodal cues does not necessarily lead to prolonged interactions. **Hypothesis 1 is not supported**.

One possible explanation is that users' willingness to engage in conversation is largely influenced by individual differences rather than the robot's adaptive capabilities. Overall, most participants (13 out of 20) showed minimal differences between conditions, with conversation rounds differing by three or fewer. However, highly engaged participants completed substantially more rounds in the adaptive condition (C2), reaching up to 54 rounds compared to 32 rounds in the random-humor condition (C1). In contrast, less engaged participants interacted very little in both conditions, completing only five or six rounds.

These observations suggest that humor adaptation alone may not be sufficient to significantly extend interactions across all users. While adaptive strategies may benefit highly engaged individuals, they appear to have little impact on those who are naturally less inclined to interact. Future research could explore the role of individual personality traits, such as extroversion or humor appreciation, in shaping responses to adaptive humor.

D. Hypothesis 2

Since the conversation was open-ended but restricted to educational topics, the learning materials varied from person to person. As a result, we could not administer a standardized quiz or test to assess learning outcomes directly. Instead, we used Q1 from the post-experiment questionnaire as a self-reported measure of learning retention to validate H2. The topics most frequently discussed by participants included Artificial Intelligence, French history, robotics, and cooking recipes, reflecting a diverse range of interests within the educational framework.

A Shapiro-Wilk test confirmed that the difference scores for all measures (Q1–Q8) between the two conditions followed a normal distribution ($p > .05$), satisfying the assumption of normality required for paired-samples t-tests. We then conducted a paired-samples t-test to compare learning retention between the two conditions. The results revealed that participants reported significantly higher learning retention in the adaptive condition (C2) (M = 7.85, SD = 1.39) compared to the random-humor condition (C1) (M = 6.75, SD = 1.59), with $t(19) = 3.688$, $p = .002$, $d = 0.82$, indicating a large effect. These findings suggest that participants perceived greater knowledge retention in the adaptive condition compared to the random-humor condition. However, as retention was measured through self-report rather than objective testing, these results should be interpreted as reflecting perceived learning outcomes rather than verified knowledge acquisition. **Hypothesis 2 is supported**.

The improvement in self-reported learning outcomes in the adaptive condition can be attributed to the cognitive and

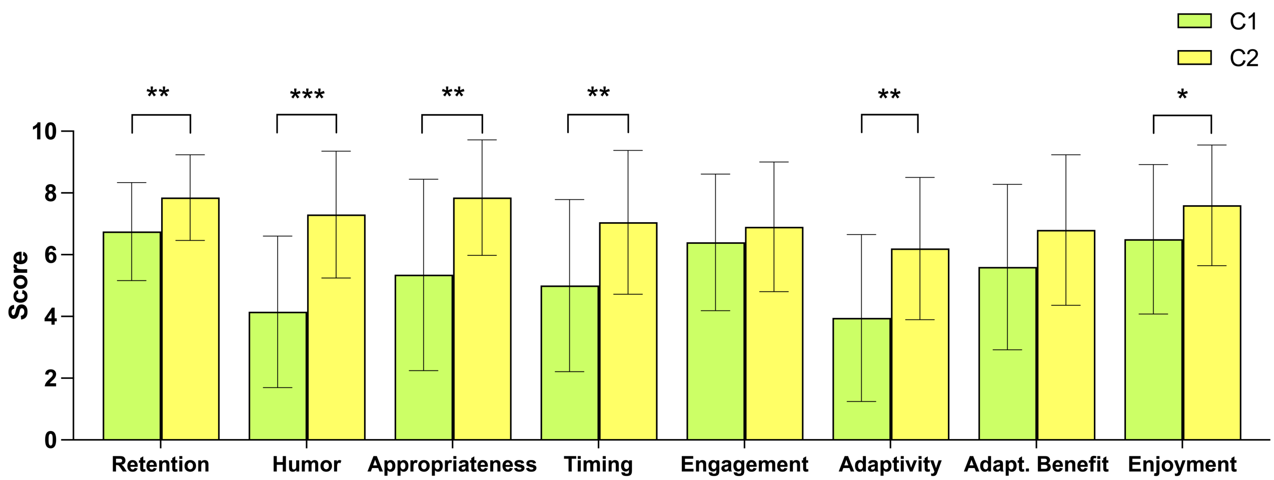

Fig. 2. Post-questionnaire results for Conditions C1 and C2 across the measured constructs. The x-axis follows the questionnaire order (Q1Q8). Error bars represent standard deviation. Significance markers (*, **, ***) indicate results from paired-samples t-tests at the 0.05, 0.01, and 0.001 levels, respectively.

emotional benefits of humor in educational settings. Prior research has demonstrated that humor can facilitate deeper information processing and enhance memory retention. For example, [35] found that students tend to recall information better when it is presented with humor, as humor creates a more engaging and less stressful learning environment. In our study, the adaptive robot's ability to integrate humor dynamically may have helped participants maintain attention, reduce cognitive load, and form stronger associations with the presented knowledge, potentially contributing to higher self-reported retention scores.

Cronbach's α for the three humor-related items (Q2, Q3, Q4) across both conditions was 0.912, indicating excellent internal consistency. Therefore, these items were averaged to form a single Humor Index (HI). Paired-samples t-tests on the individual items indicated that participants rated the adaptive robot significantly higher than the random-humor robot on humor (Q2: $t(19) = 5.688$, $p < .001$), appropriateness (Q3: $t(19) = 3.583$, $p = .002$), and timing (Q4: $t(19) = 3.297$, $p = .004$). A paired-samples t-test on the combined HI also confirmed a significant difference between the two conditions ($t(19) = 4.814$, $p < .001$).

To examine whether changes in perceived humor were associated with changes in learning outcomes, a Spearman's rank-order correlation was conducted between the differences in HI and Q1 scores ($C2 - C1$). The analysis revealed a moderate positive association ($\rho = 0.390$, $p = .013$), suggesting that participants whose perceived humor increased more in the adaptive condition also tended to show greater gains in knowledge retention. Although the effect size is modest, the correlation is statistically significant, indicating that humor perception may be associated with higher perceived retention. Individual differences and the relatively short interaction duration may limit the strength of this relationship.

Overall, these results suggest that humor, when carefully timed and appropriately integrated into educational dialogues, can foster a more engaging and comfortable learning environment, thereby supporting higher perceived knowledge retention.

E. Hypothesis 3

An overview of all post-questionnaire items is presented in Figure 2. As shown, the scores for Condition C2 are consistently higher than those for Condition C1 across all questions. However, not all differences reached statistical significance. The findings for Q1 to Q4 have been discussed in previous sections.

For H3, Q8 was predefined as the primary outcome measure capturing overall enjoyment. A paired-samples t-test showed a significant difference, with C2 (M = 7.6, SD = 1.96) rated higher than C1 (M = 6.5, SD = 2.42), $t(19) = 2.604$, $p = .017$. This indicates that the adaptive robot increased participants' overall enjoyment. **Hypothesis 3 is supported**.

For Q6, a paired-samples t-test revealed a significant difference, indicating that the perceived adaptivity of C2 (M = 6.2, SD = 2.31) was significantly higher than that of C1 (M = 3.95, SD = 2.7), with $t(19) = 3.514$, $p = .002$. This provides additional evidence that the system effectively adapts its humor and responses based on users' emotional states and engagement levels.

While other differences did not reach statistical significance, the overall trend consistently favors the adaptive condition.

VI. Conclusion

This study demonstrates the value of adaptive humor in humanrobot interaction. The proposed framework enhanced the perceived humor, appropriateness, and timing of the robot's responses, leading to higher user enjoyment and perceived information retention compared to a non-adaptive baseline. However, adaptive humor did not significantly increase conversation length, indicating that while humor enhances interaction quality, it may not be sufficient on its own to sustain prolonged engagement.

This study has several limitations. The relatively small sample size may have limited statistical power, and the reliance on self-reported learning measures may not fully reflect objective knowledge retention. In addition, the baseline condition relied on a random humor strategy, which may not represent a strong or strategically designed comparison model. Moreover, the controlled experimental setting may not capture the complexity of real-world interactions.

Future work should explore longer-term deployments, more diverse participants, objective engagement measures, and comparisons against more sophisticated baselines, while integrating richer multimodal cues and contextual dynamics, and more nuanced humor representations beyond simple intensity levels to enhance adaptive humor strategies.

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